INVESTIGATING DEEP Q-LEARNING MODIFICATIONS WITH RANDOMISED CONTROL TRIALS

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***Abstract*—Extensions to the deep Q-network (DQN) are usually evaluated with their own tuned hyperparameters and a handful of random seeds, so a reported gain may reflect the configuration rather than the extension. We evaluate four extensions, namely Double DQN, prioritised experience replay, multi-step returns and Averaged DQN, in a paired randomised controlled trial. Thirty hyperparameter configurations were sampled from a shared population, each with its own seed, and every configuration was trained with DQN and with each extension on four MinAtar games, giving 600 runs of five million frames. The primary outcome was the area under the greedy evaluation curve, compared with DQN by paired t-tests with a Bonferroni correction across all sixteen comparisons. Averaged DQN improved on DQN in Asterix, Freeway and Space Invaders, by 57.5, 20.9 and 19.5 per cent in mean AUC. Double DQN improved only in Space Invaders, by 11.4 per cent. Prioritised replay was worse in Freeway and Space Invaders, and multi-step returns were worse in Asterix, Freeway and Space Invaders, by up to 80 per cent. No extension differed detectably from DQN in Breakout. On the scale of per-configuration AUC ratios, seven of these nine findings remained significant; the gains of Averaged DQN in Asterix and of Double DQN in Space Invaders did not. Final-checkpoint endpoints agreed in direction with all nine findings but detected fewer of them, trials of three to ten configurations often missed them, and comparing the best configuration of each method reversed two of them. Under a shared training recipe, Averaged DQN was the only extension that improved on DQN in more than one game.**

***Index Terms*—deep reinforcement learning, deep Q-network, experimental design, randomised controlled trial, MinAtar, sample efficiency**

I. Introduction

The deep Q-network (DQN) of Mnih et al. [2015] learnt to play Atari 2600 games from pixels, using two stabilising devices: experience replay and a periodically updated target network. Many extensions followed, each changing one part of the algorithm. Double DQN changes how the bootstrap target is computed [van Hasselt et al. 2016], prioritised experience replay (PER) changes which transitions are replayed [Schaul et al. 2016], multi-step (n-step) returns lengthen the horizon of the target [Hernandez-Garcia and Sutton 2019; Sutton and Barto 2018], and Averaged DQN averages the estimates of several earlier target networks [Anschel et al. 2017]. Rainbow combined several such components and, in an ablation, found prioritised replay and multi-step learning to be the two most important [Hessel et al. 2018].

Each extension was, however, introduced with its own training recipe and evaluated on its own configuration. Deep reinforcement learning results are sensitive to hyperparameters, random seeds and implementation details [Henderson et al. 2018], evaluation protocols differ between papers [Machado et al. 2018], and comparisons often rest on few runs and point estimates [Colas et al. 2018; Agarwal et al. 2021]. When an extension outperforms DQN under one tuned configuration, it is therefore hard to say whether the gain belongs to the extension or to the configuration.

We address this with a randomised controlled trial in the sense of Pretorius et al. [2020]. The individuals of the trial are hyperparameter configurations sampled from a shared population, and every individual receives every treatment. Each comparison with DQN is then paired on the same configuration and seed, which removes the main effect of the configuration from the comparison while still averaging over the population. We use the MinAtar suite [Young and Tian 2019], whose small observations make 600 full training runs feasible on CPUs.

We ask two research questions. *RQ1:* across a population of configurations and under one shared training recipe, does each extension change sample efficiency, measured as the area under the evaluation curve (AUC), relative to DQN? *RQ2:* how far can the conclusions of smaller or more typical evaluations, based on a single checkpoint, a few configurations or the best configuration of each method, be trusted relative to those of the full trial? Our contributions are:

- a paired trial of four DQN extensions on 30 sampled configurations and four MinAtar games (600 runs, 3×10^9 frames), with code, raw results and provenance released[1];
- evidence that, under the shared recipe, Averaged DQN is the only extension that improves on DQN in more than one game, that the benefit of Double DQN is small and fragile, and that PER and n-step returns are harmful;

[1]https://github.com/2716924/research_project

- evidence that effects vary across the population: the learning rate changes the AUC by as much as the largest treatment effects, and in Freeway the n-step deficit is concentrated at the rate every other arm prefers;
- evidence that single-checkpoint endpoints and small trials lack power, and that comparing the best configuration of each method can reverse a conclusion.

Section II reviews background and related work, Section III describes the trial, Section IV reports the results, Section V discusses them and Section VI states the limitations.

II. Background and Related Work

A. Deep Q-Networks

An agent interacting with a Markov decision process observes a state s_t, takes an action a_t, receives a reward r_{t+1} and moves to s_{t+1}; it seeks to maximise the expected return discounted by $\gamma \in [0, 1)$ [Sutton and Barto 2018]. Q-learning [Watkins and Dayan 1992] estimates the optimal action-value function by bootstrapping from $\max_{a'} Q(s', a')$. DQN [Mnih et al. 2015] approximates Q with a neural network with parameters θ, stores transitions in a replay buffer $\mathcal{D}$ [Lin 1992] and minimises

$$L(\theta) = \mathbb{E}_{(s,a,r,s') \sim U(\mathcal{D})} \big[(y - Q(s, a; \theta))^2 \big] \tag{1}$$

with the target

$$y = r + \gamma \max_{a'} Q(s', a'; \theta^-), \tag{2}$$

where θ^- are the parameters of a target network copied from θ every C steps, and $y = r$ when s' is terminal.

B. The Four Extensions

Double DQN [van Hasselt et al. 2016]. The maximum in (2) uses the same estimates to select and to evaluate the next action, which biases the target upwards. Double DQN selects the action with the online network and evaluates it with the target network:

$$y = r + \gamma \, Q\big(s', \arg\max_{a'} Q(s', a'; \theta); \theta^-\big). \tag{3}$$

Prioritised experience replay [Schaul et al. 2016]. Instead of sampling uniformly, PER samples transition i with probability

$$P(i) = \frac{p_i^\alpha}{\sum_k p_k^\alpha}, \qquad p_i = |\delta_i| + \epsilon, \tag{4}$$

where δ_i is the latest temporal-difference (TD) error of the transition. New transitions receive the largest priority seen so far. Importance-sampling weights $w_i = (|\mathcal{D}| \, P(i))^{-\beta} / \max_j w_j$ correct the bias of non-uniform sampling, with β annealed linearly to 1. For the proportional variant used here, Schaul et al. [2016] chose $\alpha = 0.6$ and $\beta_0 = 0.4$ by a coarse grid search on eight games, with Double DQN rather than DQN as the base agent, and they reduced the step size by a factor of four relative to that agent because prioritised replay picks high-error transitions more often, which makes the typical gradient larger.

Multi-step returns [Hernandez-Garcia and Sutton 2019; Sutton and Barto 2018]. The n-step target sums n observed rewards before bootstrapping:

$$y = \sum_{k=0}^{n-1} \gamma^k r_{t+k+1} + \gamma^n \max_{a'} Q(s_{t+n}, a'; \theta^-). \tag{5}$$

Without an off-policy correction, as in Rainbow [Hessel et al. 2018], the reward sum follows the exploratory behaviour policy rather than the greedy policy, which trades bias for faster propagation of reward.

Averaged DQN [Anschel et al. 2017]. Averaged DQN keeps the last K target networks $\theta_1^-, \ldots, \theta_K^-$ and averages their action values before taking the maximum:

$$y = r + \gamma \max_{a'} \frac{1}{K} \sum_{k=1}^{K} Q(s', a'; \theta_k^-). \tag{6}$$

This reduces the variance of the target approximation error at the cost of K forward passes per target, with the same number of gradient updates as DQN [Anschel et al. 2017].

C. Component Studies

Rainbow's ablation on 57 Atari games found that removing prioritised replay or multi-step learning caused the largest drops in median performance [Hessel et al. 2018]. Obando-Ceron and Castro [2021] repeated the additive and ablative studies on smaller environments, including the five MinAtar games, and found multi-step returns to be the component that most improved DQN on MinAtar. Their agents used the Adam optimiser [Kingma and Ba 2015] and fixed hyperparameters apart from a learning rate swept for each component, with 10 runs per setting. With Adam, the mean squared error (MSE) loss gave much better results than the Huber loss in every environment they tested, and on Atari Adam with the MSE loss was overwhelmingly better than RMSProp with the Huber loss; with RMSProp, however, the Huber loss tended to perform better than the MSE. Fedus et al. [2020] found that uncorrected n-step returns benefit uniquely from a larger replay capacity, whereas DQN does not. Fujimoto et al. [2020] showed that a loss evaluated on non-uniformly sampled data can be rewritten as a uniformly sampled loss with the same expected gradient, which ties the effect of PER to the loss it is combined with.

D. Evaluation Methodology

Henderson et al. [2018] showed that hyperparameter settings, random seeds, environment properties and codebases can each change reported performance substantially. Machado et al. [2018] revisited the evaluation protocols of the Arcade Learning Environment and introduced sticky actions to make it stochastic. Colas et al. [2018] related the number of random seeds to the error rates of statistical tests, and Agarwal et al. [2021] showed that conclusions drawn from point estimates over few runs can differ substantially from those of a more thorough statistical analysis, and advocated interval estimates and the probability of improvement. Jordan et al. [2020]

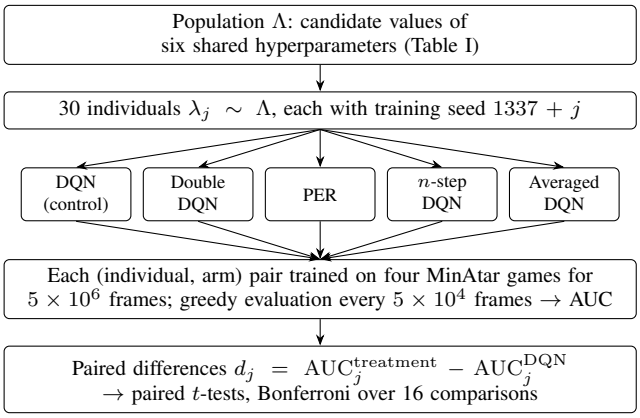


Fig. 1. Design of the trial. Every individual (a hyperparameter configuration with its own seed) is trained under all five arms, so each treatment is compared with DQN on the same configuration.

TABLE I
SHARED HYPERPARAMETER POPULATION AND THE SAMPLED INDIVIDUALS

Hyperparameter	Values (individuals in sample)
Learning rate	10^{-4} (7), $5{\times}10^{-4}$ (13), 10^{-3} (10)
Replay buffer capacity	10^4 (15), $5{\times}10^4$ (5), 10^5 (10)
Mini-batch size	32 (15), 64 (15)
Target update period (steps)	500 (8), 1000 (11), 2000 (11)
ε-decay duration (steps)	$5{\times}10^4$ (11), 10^5 (7), $2{\times}10^5$ (12)
Hidden layers	1 (16), 2 (14)

argued that tuning hyperparameters and then reporting performance answers the wrong question, and evaluated algorithms made 'complete' by sampling their hyperparameters at random from a fixed distribution in every trial. Patterson et al. [2024] give a broader guide to empirical design in RL, including hyperparameter sensitivity, hypothesis testing and experimenter bias. Pretorius et al. [2020] brought the randomised controlled trial to deep learning: they sampled network designs from a population, trained identical copies of each design under every intervention and tested the differences, over more than 12,000 trained networks. We apply this design to DQN extensions: like Jordan et al., we sample configurations at random, but every arm is trained on the same draws, so each comparison is paired.

III. METHOD

A. Trial Design

Fig. 1 summarises the design. The population Λ is the Cartesian product of the candidate values of six shared hyperparameters (Table I). Each of the $N = 30$ individuals λ_j was drawn by sampling every hyperparameter independently and uniformly from its candidates with a seeded generator, and was given its own training seed $1337 + j$. This is a simple random sample, and Table I shows the resulting counts, which are unbalanced for the replay buffer capacity. Every run filled the buffer with 5,000 transitions, collected by uniformly random actions, before learning began.

TABLE II
TRAINING SETTINGS

Setting	Value
Network	convolution (16 filters, 3×3, stride 1), then 1 or 2 hidden layers of 128 units (ReLU)
Optimiser, loss	RMSProp (decay 0.99, $\epsilon = 10^{-8}$, no momentum); MSE
Discount, exploration	$\gamma = 0.99$; ε from 1.0 to 0.1, linear over the sampled duration
Updates	one per step after 5,000 random transitions
Training length	5×10^6 frames (no frame skipping)
Evaluation	every 5×10^4 frames; 10 greedy episodes with seeds 1000–1009
PER	proportional; $\alpha = 0.6$; β from 0.4 to 1 over training; $\epsilon = 10^{-6}$
n-step DQN	$n = 3$; no off-policy correction
Averaged DQN	$K = 10$ target networks, one stored at each target update

Five arms were trained on every individual: DQN, the control, and four treatments, each of which changes one component (Section II-B). The treatment-specific settings were fixed rather than sampled: $\alpha = 0.6$ and $\beta_0 = 0.4$ for PER [Schaul et al. 2016], with the small priority constant $\epsilon = 10^{-6}$ our own choice; $n = 3$ for n-step DQN [Hessel et al. 2018]; and $K = 10$ for Averaged DQN, one of the values evaluated by Anschel et al. [2017]. Within an individual, all arms share the configuration, the training seed and the evaluation seeds, so every comparison with DQN is paired.

The arms were trained on four MinAtar games [Young and Tian 2019]: Breakout, Asterix, Freeway and Space Invaders (version 1, with the minimal action set). MinAtar's default settings were kept, including sticky actions with probability 0.1 [Young and Tian 2019] and difficulty ramping in the games that support it. Seaquest was also run but is excluded: in each of the four arms in which it was run, a greedy evaluation episode exceeded the 100,000-step limit, because MinAtar never ends a Seaquest episode in which the submarine stays on the surface, where it starts. No run in the four reported games was aborted. The trial therefore comprises $4 \times 5 \times 30 = 600$ runs.

B. Agent and Shared Training Recipe

Table II lists the training settings. All arms use the network of the MinAtar DQN baseline [Young and Tian 2019], whose 16 filters and 128 hidden units are a quarter of the corresponding layers of DQN [Mnih et al. 2015]; whether there are one or two hidden layers is sampled. Training uses RMSProp [Tieleman and Hinton 2012] with the MSE loss (1). MinAtar has no frame skipping, so a frame is one environment step, and each run lasts 5×10^6 frames, as in the MinAtar baselines.

The recipe is deliberately shared and is not the recipe of any one paper: DQN is usually trained with the Huber loss [Obando-Ceron and Castro 2021], and Schaul et al. [2016] lowered the step size for PER. Changing one component under a fixed recipe is what isolates its effect, but it also means that the results concern these components under this recipe.

The treatments were implemented as follows. PER stores priorities in a sum tree, draws each mini-batch by stratified sampling over equal segments of the total priority, normalises the importance weights by the largest possible weight, anneals β linearly over the 5×10^6 frames and averages the weighted squared TD errors. n-step DQN stores, for each transition, the discounted sum of the next three rewards and the state three steps later; at the end of an episode, the remaining shorter returns are stored without a bootstrap. Averaged DQN stores a copy of the target network at every target update and averages the last ten copies as in (6), or all of them while fewer than ten exist.

C. Evaluation and Primary Outcome

Every 5×10^4 frames, the online network was evaluated greedily ($\varepsilon = 0$) for 10 episodes, with the same environment seeds (1000–1009) for every individual, arm and checkpoint. The mean return G_i at checkpoint i is one point on the evaluation curve, giving 100 points per run. An evaluation episode longer than 10^5 steps aborts the run without recording a score. The primary outcome is the trapezoidal area under the curve,

$$\mathrm{AUC} = \sum_{i=1}^{99} \frac{G_i + G_{i+1}}{2}\,\Delta x, \qquad \Delta x = 5 \times 10^4 \text{ frames}, \quad (7)$$

reported in units of 10^6 (return $\times$ frames). The AUC rewards both fast learning and a high final level: divided by the 4.95×10^6 frames it spans, it is the average evaluation return over training.

D. Statistical Analysis

The statistical unit is the individual; the ten evaluation episodes and the 100 checkpoints are repeated measurements within a run, not replicates. For each game and treatment, we form the 30 paired differences $d_j = \mathrm{AUC}_j^{\mathrm{treatment}} - \mathrm{AUC}_j^{\mathrm{DQN}}$ and test whether their mean is zero with a two-sided paired t-test. Because our conclusions are drawn across games, we treat all $4 \times 4 = 16$ comparisons as one family and apply the Bonferroni correction [Dunn 1961], $\alpha = 0.05/16 = 0.003125$, with matching simultaneous 99.69% t-intervals. We report each effect as the mean difference as a percentage of the DQN mean, with the standardised effect $d_z = \bar{d}/s_d$ [Lakens 2013] and the number of individuals for which the treatment beat DQN. We report the t-test as primary and the Wilcoxon signed-rank test [Wilcoxon 1945] as a sensitivity analysis for every comparison, rather than choosing between them with a normality pre-test, which alters the error rates of the test that follows [Rochon et al. 2012]; Shapiro–Wilk [Shapiro and Wilk 1965] rejected normality of the differences at the 0.05 level for 6 of the 16 comparisons.

A percentage of the DQN mean treats that mean as known, and its interval can extend below -100% although the AUC is positive. We therefore also analyse the per-individual log ratios $\log(\mathrm{AUC}_j^{\mathrm{treatment}}/\mathrm{AUC}_j^{\mathrm{DQN}})$: their back-transformed mean is the geometric-mean ratio, which we report with the same 99.69% t-interval and a one-sample t-test (p_r).

Two secondary endpoints address RQ2: the mean of the last ten evaluations (the final 5×10^5 frames) and the final evaluation alone. The fraction of individuals for which a treatment beats DQN estimates the probability that it does so on a randomly drawn configuration, a paired analogue of the probability of improvement of Agarwal et al. [2021]; we give 95% Wilson intervals [Wilson 1927]. To mimic smaller trials, we repeat each significant comparison on every subset of $k \in \{3, 5\}$ individuals and on 10^5 random subsets of $k = 10$, and record how often a paired t-test at $\alpha = 0.05$ detects the effect in the direction of the full trial. To mimic separately tuned methods, we compare the best AUC of each arm over the 30 configurations with that of DQN, a comparison that is biased upwards and only illustrative. Finally, as the six hyperparameters were sampled independently, we regress log AUC on all six, as categorical factors, for each arm and game. The bands in the learning curves are pointwise 95% bootstrap intervals over individuals (2,000 resamples, Efron 1979). All analyses other than the Bonferroni-corrected paired t-tests on the AUC are exploratory.

E. Implementation

The agents were implemented in PyTorch [Paszke et al. 2019] (version 2.13) with the Gymnasium [Towers et al. 2024] interface to MinAtar (version 1.0.15) and trained on CPUs, with four threads per job, on the MSL cluster at the University of the Witwatersrand. The code includes unit tests and analytic checks of every arm's target and replay computations, which run before each cluster job. Among them, PER's sampling probabilities and importance weights are checked against analytic values, n-step returns against an independent episode-level reference, and n-step DQN with $n = 1$ is checked to reproduce the network weights of DQN exactly through the MinAtar training loop.

IV. Results

A. Primary Outcome

Table III and Fig. 2 report the 16 comparisons, nine of which are significant at the family-wide threshold. Averaged DQN had a higher AUC than DQN in Asterix ($+57.5\%$), Freeway ($+20.9\%$) and Space Invaders ($+19.5\%$), and beat DQN on 25, 24 and 26 of the 30 individuals. Double DQN was better only in Space Invaders ($+11.4\%$, 21 of 30 individuals); its differences in Asterix ($+28.2\%$, $p_t = 0.048$) and Freeway (-8.5%, $p_t = 0.014$) were not significant. PER was worse in Freeway (-35.3%) and Space Invaders (-50.0%), beating DQN on only 3 and 1 individuals. n-step DQN was worse in Asterix (-80.0%), Freeway (-59.4%) and Space Invaders (-21.0%), with 2, 0 and 6 wins. The significant effects have $|d_z|$ between 0.59 and 1.64. Holm's step-down procedure [Holm 1979], which is uniformly at least as powerful as the Bonferroni correction, rejects the same nine hypotheses: the tenth-smallest p-value, 0.014, exceeds $0.05/7$.

On the ratio scale, seven of the nine findings remain significant at the family-wide threshold (Table III, Fig. 2). The two exceptions are gains. Averaged DQN in Asterix has

TABLE III
MEAN AUC (10^6, STANDARD DEVIATION IN PARENTHESES) AND PAIRED COMPARISONS WITH DQN ($n = 30$ PER ARM)

Game	Arm	AUC	Δ (%)	Ratio (99.69% CI)	d_z	p_t	p_W	p_r
Breakout	DQN	65.2 (16.6)						
	Double DQN	65.3 (16.9)	+0.1	1.00 [0.93, 1.08]	+0.01	0.94	0.92	0.94
	PER	66.7 (18.3)	+2.3	1.01 [0.90, 1.15]	+0.10	0.58	0.36	0.70
	n-step DQN	68.1 (20.0)	+4.5	1.03 [0.89, 1.18]	+0.20	0.27	0.24	0.58
	Averaged DQN	65.9 (17.6)	+1.1	1.01 [0.91, 1.12]	+0.07	0.72	0.72	0.82
Asterix	DQN	16.8 (14.1)						
	Double DQN	21.5 (21.7)	+28.2	1.04 [0.69, 1.58]	+0.38	0.048	0.13	0.76
	PER	14.0 (9.6)	−16.5	0.98 [0.68, 1.41]	−0.31	0.10	0.27	0.83
	n-step DQN	3.3 (2.4)	−80.0	0.29 [0.15, 0.57]	−0.92	$\mathbf{2.3 \times 10^{-5}}$	$\mathbf{1.2 \times 10^{-6}}$	$\mathbf{1.8 \times 10^{-6}}$
	Averaged DQN	26.4 (25.1)	+57.5	1.38 [0.89, 2.12]	+0.63	**0.0017**	$\mathbf{4.2 \times 10^{-4}}$	0.024
Freeway	DQN	159.8 (58.9)						
	Double DQN	146.2 (64.8)	−8.5	0.89 [0.78, 1.02]	−0.48	0.014	0.019	0.0083
	PER	103.4 (63.7)	−35.3	0.58 [0.41, 0.82]	−1.03	$\mathbf{4.1 \times 10^{-6}}$	$\mathbf{1.0 \times 10^{-7}}$	$\mathbf{2.1 \times 10^{-5}}$
	n-step DQN	64.9 (54.9)	−59.4	0.31 [0.17, 0.57]	−1.31	$\mathbf{6.7 \times 10^{-8}}$	$\mathbf{1.9 \times 10^{-9}}$	$\mathbf{8.3 \times 10^{-7}}$
	Averaged DQN	193.2 (52.4)	+20.9	1.25 [1.07, 1.47]	+0.99	$\mathbf{7.7 \times 10^{-6}}$	$\mathbf{1.1 \times 10^{-5}}$	$\mathbf{8.8 \times 10^{-5}}$
Space Invaders	DQN	168.8 (90.3)						
	Double DQN	188.0 (100.6)	+11.4	1.10 [0.97, 1.26]	+0.59	**0.0029**	0.012	0.023
	PER	84.4 (65.4)	−50.0	0.43 [0.30, 0.60]	−1.64	$\mathbf{7.1 \times 10^{-10}}$	$\mathbf{9.3 \times 10^{-9}}$	$\mathbf{1.2 \times 10^{-8}}$
	n-step DQN	133.4 (106.0)	−21.0	0.74 [0.61, 0.88]	−0.72	$\mathbf{4.7 \times 10^{-4}}$	$\mathbf{1.9 \times 10^{-4}}$	$\mathbf{9.3 \times 10^{-6}}$
	Averaged DQN	201.7 (111.0)	+19.5	1.18 [1.05, 1.33]	+1.03	$\mathbf{4.1 \times 10^{-6}}$	$\mathbf{3.8 \times 10^{-6}}$	$\mathbf{1.3 \times 10^{-4}}$

Δ: mean paired difference as a percentage of the DQN mean. Ratio: geometric mean of the per-individual ratio of the treatment's AUC to that of DQN, with its Bonferroni-adjusted 99.69% t-interval on the log scale. d_z: mean difference divided by the standard deviation of the differences. p_t, p_W: two-sided paired t-test and Wilcoxon signed-rank p-values (primary analysis); p_r: one-sample t-test of the log ratios. Bold values are at most $0.05/16$.

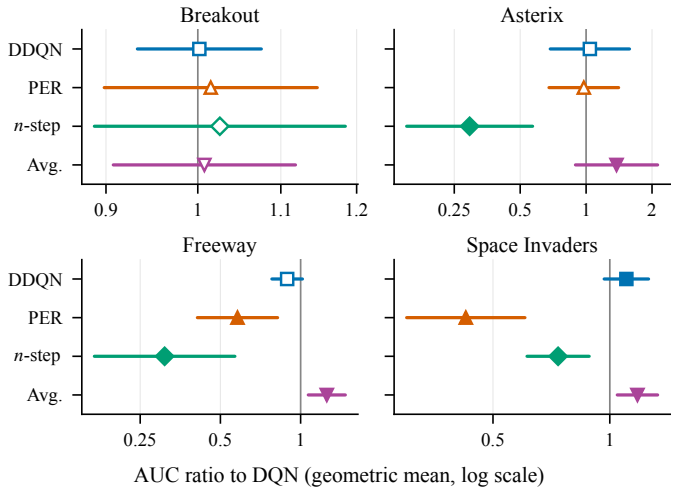


Fig. 2. Geometric-mean AUC ratio of each treatment to DQN, on a log scale, with Bonferroni-adjusted 99.69% intervals. Filled markers: significant in the primary paired t-test ($p_t \leq 0.05/16$). The horizontal scales differ between games. DDQN: Double DQN; Avg.: Averaged DQN.

a geometric-mean ratio of 1.38 $[0.89, 2.12]$ ($p_r = 0.024$): it beat DQN on 25 individuals, but on two others it reached only 0.10 and 0.28 of the AUC of DQN, and these dominate on the log scale; a Wilcoxon test on the log ratios still rejects ($p = 0.0016$). Double DQN in Space Invaders has a ratio of 1.10 $[0.97, 1.26]$ ($p_r = 0.023$). The log ratios were not closer to normal than the differences: Shapiro–Wilk rejected normality for 9 of the 16 comparisons.

In Breakout no treatment differed significantly from DQN. The ratios lie between 1.00 and 1.03, but their intervals extend from 0.89 to 1.18, so the data do not show that the extensions are equivalent to DQN there either.

B. Learning Dynamics

The learning curves (Fig. 3) show how these differences arise. In Breakout the curves rise quickly and then overlap for the rest of training, ending between 13.2 and 14.8 (mean of the last ten evaluations); n-step DQN learnt fastest early on, with a mean return of 14.0 against 12.0 for DQN at 5×10^5 frames. In Asterix, n-step DQN barely improved: its mean curve never exceeded 1.24, whereas DQN ended at 3.8. In Freeway, Averaged DQN led from early in training (31.6 against 20.1 for DQN at 5×10^5 frames) and ended highest (44.1 against 38.4). In Space Invaders, PER started level with DQN (23.6 against 23.5 at the first evaluation), fell to 13.0 by 10^6 frames and ended at about half the level of DQN (19.1 against 35.9).

C. Variation Across Configurations

Performance varied widely between configurations (Fig. 4). For DQN, the standard deviation of the AUC across individuals ranged from 26% (Breakout) to 84% (Asterix) of its mean. The paired design removes the main effect of the configuration from the comparisons; the interaction between configuration and treatment remains in the variance of the differences. Several of the distributions are skewed or have more than one mode. The most extreme are those of n-step DQN: in Asterix, 29 of its 30 individuals had an AUC of at most 5, and in Freeway its seven individuals at the lowest learning rate all had an AUC between 8.2 and 28.0, against 144.8 to 231.9 for DQN at that rate.

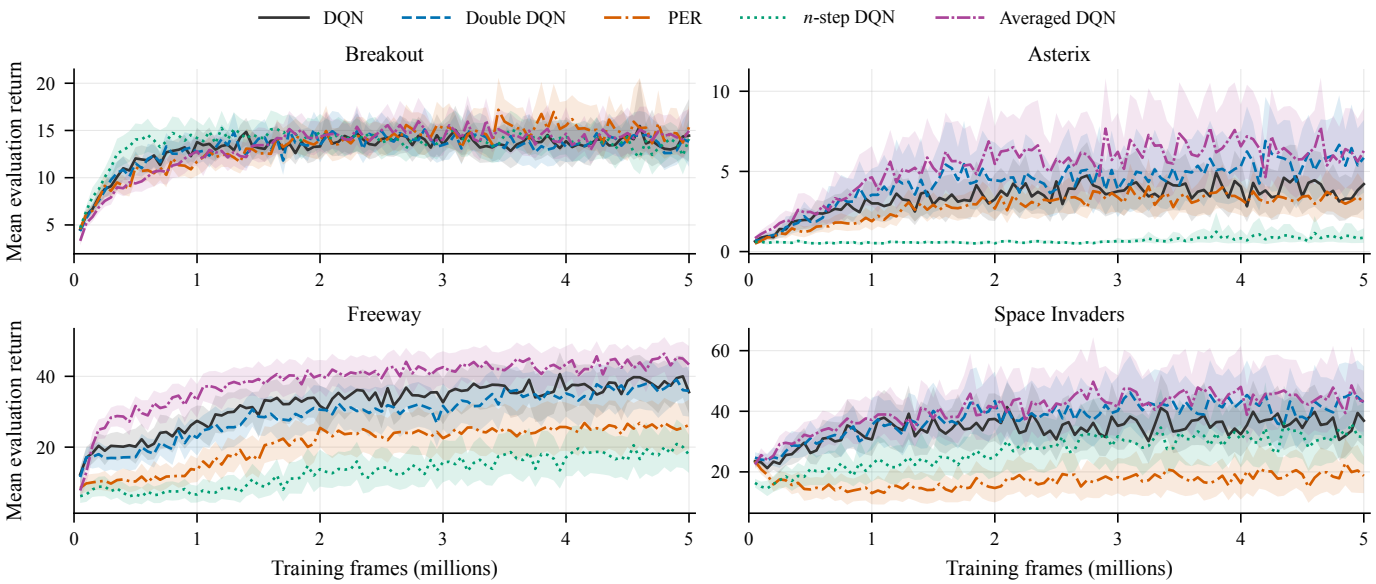


Fig. 3. Mean greedy evaluation return during training ($n = 30$ individuals per arm). Shaded bands are pointwise 95% bootstrap intervals over individuals; the curves are not smoothed.

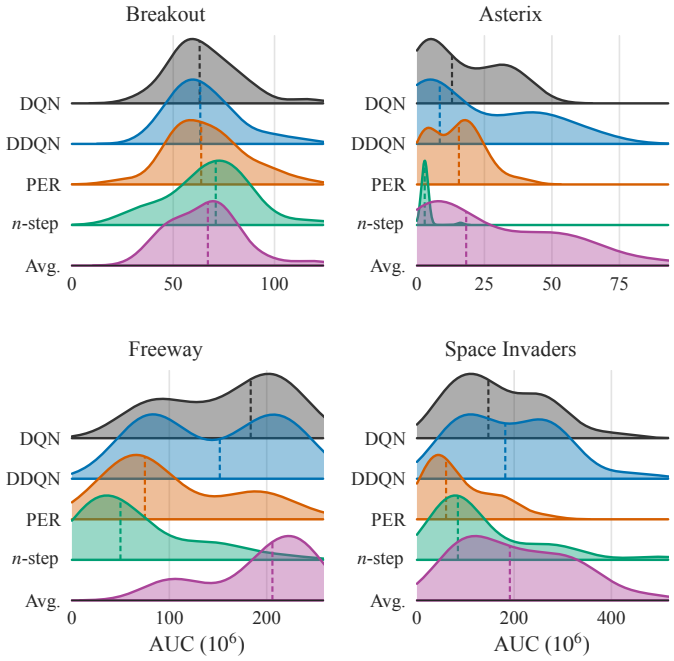


Fig. 4. Distribution of the AUC over the 30 individuals of each arm, in the style of Pretorius et al. [2020]: Gaussian kernel density estimates (Scott's bandwidth, clipped at zero), each scaled to its own peak height. Dashed lines mark the medians. DDQN: Double DQN; Avg.: Averaged DQN.

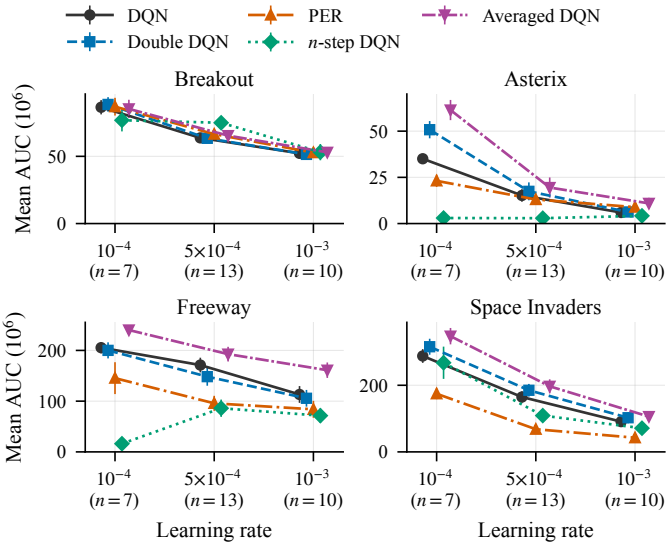


Fig. 5. Mean AUC by sampled learning rate, with ± 1 standard error over the n individuals at each rate. The subgroups also differ in their other hyperparameters, so the comparison is descriptive; Table VI adjusts for them.

Much of this variation was associated with the learning rate (Fig. 5). The mean AUC of DQN at a learning rate of 10^{-4} was 1.7 to 5.9 times its mean at 10^{-3} (Breakout 86.5 against 52.2, Asterix 35.1 against 6.0, Freeway 205.4 against 113.2, Space Invaders 287.9 against 90.8). Because these subgroups also differ in their other hyperparameters, we regressed log AUC on all six sampled factors (Table VI, Appendix B). Adjusted for the other five, raising the learning rate from 10^{-4} to 10^{-3} multiplied the AUC of DQN by 0.21 (Asterix) to 0.57 (Freeway), and the learning rate explained 58% to 93% of the variance in log AUC that the other factors left unexplained (partial R^2). On this common log scale, the adjusted learning-rate effect on DQN was larger than the geometric-mean treatment effect in 15 of the 16 comparisons; the exception was n-step DQN in Freeway. The lowest learning rate gave the highest mean AUC for every arm in every game except n-step DQN in Asterix and Freeway.

The two n-step deficits responded differently to the learning

rate. In Asterix, n-step DQN barely learnt at any rate: its mean AUC was 3.0, 2.9 and 4.2 from the lowest rate to the highest, and its ratio of mean AUC to that of DQN rose from 0.09 to 0.70 only because DQN did worse at higher rates. In Freeway, by contrast, n-step DQN failed at 10^{-4}, the rate at which every other arm did best: its mean AUC there was 16.1, against 205.4 for DQN, whereas at the two larger rates it reached 86.1 and 71.4 (ratios of means to DQN of 0.50 and 0.63). Adjusted for the other factors, its AUC at 10^{-3} was 4.12 $[2.31, 7.34]$ times that at 10^{-4}, whereas the corresponding factor was below one for DQN, Double DQN and Averaged DQN. PER stayed below DQN at every learning rate in Freeway and Space Invaders, with ratios of means to DQN between 0.41 and 0.74 and no consistent trend; at 10^{-4} the ratios were 0.71 and 0.61.

The gain of Averaged DQN did not shrink consistently at longer target-update periods: at periods of 500, 1000 and 2000 steps (8, 11 and 11 individuals), its geometric-mean ratio to DQN was 1.23, 1.25 and 1.27 in Freeway, 0.85, 1.78 and 1.51 in Asterix and 1.21, 1.26 and 1.08 in Space Invaders. In the regression, a period of 2000 rather than 500 steps changed the AUC of DQN itself by a factor of 0.83 to 1.12.

D. Robustness

The nine findings survive several checks. With a Bonferroni correction within each game instead ($\alpha = 0.05/4 = 0.0125$), the same nine comparisons are significant. The Wilcoxon test confirms eight of them at the family-wide threshold; the exception is Double DQN in Space Invaders ($p_W = 0.012$), whose interval also only just excludes zero ($[+0.1\%, +22.6\%]$), so this finding rests on the t-test alone.

The secondary endpoints (Table V) agree in direction with all nine findings, but they are noisier. With the mean of the last ten evaluations, five of the nine remain significant at the family-wide threshold; with the final evaluation alone, only three do. No comparison that was not significant on the AUC became significant on either secondary endpoint. The single-checkpoint endpoints therefore have less power than the AUC; they do not point to different conclusions.

E. Reliability of Smaller Evaluations

Table IV shows how smaller or more typical evaluations would have fared. The paired probability of improvement of Averaged DQN was 0.83, 0.80 and 0.87 in Asterix, Freeway and Space Invaders, so a comparison on one configuration drawn at random would have favoured DQN with a probability of 0.13 to 0.20. Trials of a few configurations usually got the direction of a finding right: the mean difference had the sign of the full trial in 84% to 100% of the subsets of three individuals and 92% to 100% of those of five. They seldom reached significance, however: a t-test detected the effects in 6% to 30% of the subsets of three, 9% to 79% of those of five and 37% to 100% of those of ten, and detected the two weakest findings, Double DQN in Space Invaders and Averaged DQN in Asterix, in fewer than half of the subsets of ten. Significance in the wrong direction occurred in at most 0.4% of subsets.

TABLE IV
RELIABILITY OF SMALLER EVALUATIONS

		Best of 30	Detected (%)		
Arm	PoI (95% CI)	Δ (%)	$k = 3$	5	10
Breakout					
DDQN	0.47 [0.30, 0.64]	−1.5	–	–	–
PER	0.57 [0.39, 0.73]	−2.5	–	–	–
n-step	0.63 [0.46, 0.78]	+3.3	–	–	–
Avg.	0.57 [0.39, 0.73]	+2.9	–	–	–
Asterix					
DDQN	0.57 [0.39, 0.73]	+50.8	–	–	–
PER	0.47 [0.30, 0.64]	−10.9	–	–	–
n-step	0.07 [0.02, 0.21]	−62.9	8	20	80
Avg.	0.83 [0.66, 0.93]	+106.1	7	14	42
Freeway					
DDQN	0.33 [0.19, 0.51]	−2.7	–	–	–
PER	0.10 [0.03, 0.26]	+2.0	13	33	98
n-step	0.00 [0.00, 0.11]	−8.4	14	54	100
Avg.	0.80 [0.63, 0.90]	+4.7	13	39	83
Space Invaders					
DDQN	0.70 [0.52, 0.83]	+13.0	6	9	37
PER	0.03 [0.01, 0.17]	−35.7	30	79	100
n-step	0.20 [0.10, 0.37]	+24.7	10	24	55
Avg.	0.87 [0.70, 0.95]	+18.4	17	42	86

PoI: fraction of individuals for which the treatment's AUC exceeds that of DQN, with 95% Wilson interval. Best of 30: best AUC of the arm over the 30 configurations relative to the best of DQN. Detected: percentage of subsets of k individuals in which a paired t-test at $\alpha = 0.05$ is significant in the direction of the full trial (all subsets for $k = 3, 5$; 10^5 random subsets for $k = 10$), shown for the nine significant comparisons.

Comparing the best configuration of each arm with that of DQN, as a study that tunes each method separately would, reversed two findings: n-step DQN was 24.7% better than DQN in Space Invaders and PER 2.0% better in Freeway.

V. DISCUSSION

A. RQ1: Which Extensions Help?

Averaged DQN was the only extension that improved on DQN in more than one game, on at least 24 of 30 individuals in each of three games, and the Wilcoxon test confirmed all three; on the ratio scale its gain in Asterix was not significant. It was designed to reduce the variance of the target [Anschel et al. 2017], which we did not measure. Averaging ten target networks also makes the target depend on older parameters; if that were the cause, the gain should shrink at long target-update periods and long periods should help DQN, but neither happened consistently (Section IV-C).

Double DQN helped little. Its one significant result, in Space Invaders, was the weakest of the nine findings: neither the Wilcoxon test nor the ratio scale confirmed it, and its other estimates had mixed signs. We did not measure overestimation, so we cannot tell whether it was small under this recipe or whether reducing it did not raise the AUC.

PER and n-step returns were harmful, unlike in the Rainbow ablation on Atari [Hessel et al. 2018] and, for n-step returns, the MinAtar study of Obando-Ceron and Castro [2021]. Is this the recipe or an implementation error? The analytic checks of every arm (Section III-E) make a gross error unlikely, but they cannot exclude an interaction between our implementation and the recipe. Without a positive control, such as DQN against

PER and n-step DQN under the recipe of Obando-Ceron and Castro, the explanations below are hypotheses.

Schaul et al. [2016] reduced the step size for PER, and Obando-Ceron and Castro used Adam with one update every four steps, whereas we used RMSProp with the MSE loss and one update per step; with RMSProp, they found that the Huber loss tended to work better than the MSE on Atari, although Adam with the MSE loss was their best combination. A step size too large for PER should matter least at the lowest learning rate, yet PER remained 29% and 39% below DQN at 10^{-4} in Freeway and Space Invaders, so the step size alone does not explain its deficit. Because non-uniform sampling has the same expected gradient as a different loss under uniform sampling [Fujimoto et al. 2020], PER with the MSE may weight large errors more heavily than intended; recording the importance weights and TD errors during training would allow this to be examined.

The two n-step deficits differ in kind (Section IV-C). In Asterix, n-step DQN barely learnt at any learning rate. In Freeway, it failed at the rate at which every other arm did best and partly recovered at larger rates, an interaction that a study tuning each method separately would not see. We found no consistent association with the replay buffer capacity, the factor that Fedus et al. [2020] identified for n-step returns, but the subgroups are small.

B. RQ2: How Reliable Are Smaller Evaluations?

Evaluating whole learning curves over a population of configurations exposes how unreliable smaller or more typical evaluations are (Section IV-E). The single-checkpoint endpoints differed from the AUC in power, not in conclusions: they agreed in direction with all nine findings but detected only three or five. Small trials usually pointed the right way but seldom reached significance, so the absence of a significant effect in a small evaluation says little. Comparing the best configuration of each method, as separately tuned studies effectively do, reversed two findings, because the effects depend on the configuration.

The paired design makes the effects detectable despite this dependence: the significant effects have $|d_z|$ between 0.59 and 1.64, whereas the spread of the AUC between individuals is often as large as the differences between arms (Fig. 4).

Isolated under a shared recipe, the effects only partly matched those reported in the literature: the benefits reported for PER and n-step returns did not appear, whereas those of Averaged DQN did.

VI. Limitations

Each configuration was trained with one seed, so within an individual the effects of the configuration and the seed cannot be separated; both are shared across arms, so the comparisons remain paired. The results hold for one shared recipe with treatment settings fixed at published values; tuning each arm separately could change them, and without a positive control under another recipe an interaction between our implementation and the recipe cannot be fully excluded. The grid of learning rates does not bracket the optimum: the lowest rate was best for almost every arm, so the treatment effects are conditional on a population in which 23 of the 30 configurations probably use too large a learning rate. The trial covers four MinAtar games and 5×10^6 frames. Seaquest was lost to a protocol choice: the evaluation aborted a run whose greedy episode exceeded 10^5 steps instead of capping the episode, which would have kept the game in the trial. The subgroup analyses are descriptive, and all analyses beyond the primary comparisons are exploratory (Section III-D).

VII. Conclusion

In a paired randomised controlled trial of four DQN extensions over 30 sampled configurations and four MinAtar games, only Averaged DQN improved on DQN in more than one game. Double DQN helped only slightly and not robustly, and prioritised replay and multi-step returns were harmful under the shared recipe, contrary to earlier component studies. The learning rate changed performance by as much as the largest treatment effects, small evaluations detected fewer of the findings, and comparing the best configuration of each method reversed two of them. Future work should run a positive control under the recipe of Obando-Ceron and Castro [2021], extend the learning-rate grid downwards, add seeds per configuration, and evaluate Seaquest with a capped protocol.

Acknowledgement

I express my sincere gratitude to my supervisors, Steven James and Benjamin Rosman, for their invaluable guidance, patience, and continuous support throughout the formulation of this research report.

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APPENDIX A
ADDITIONAL RESULTS

Table V gives the paired comparisons on the two secondary endpoints.

TABLE V
SECONDARY ENDPOINTS: PAIRED COMPARISONS WITH DQN

		Last ten evaluations		Final evaluation	
Game	Arm	Δ (%)	p_t	Δ (%)	p_t
Breakout	DDQN	-0.8	0.87	-4.4	0.65
	PER	$+7.0$	0.40	$+6.2$	0.62
	n-step	-4.6	0.42	-2.5	0.79
	Avg.	$+4.8$	0.31	$+3.5$	0.69
Asterix	DDQN	$+53.9$	0.0071	$+39.0$	0.060
	PER	-14.0	0.20	-27.0	0.058
	n-step	-76.4	$\mathbf{6.3 \times 10^{-4}}$	-79.3	$\mathbf{0.0013}$
	Avg.	$+62.4$	$\mathbf{0.0023}$	$+49.3$	0.020
Freeway	DDQN	-3.4	0.58	$+1.0$	0.92
	PER	-32.7	0.0032	-26.1	0.029
	n-step	-50.6	$\mathbf{3.1 \times 10^{-5}}$	-49.9	$\mathbf{2.0 \times 10^{-4}}$
	Avg.	$+14.8$	0.012	$+21.9$	0.0092
Space Invaders	DDQN	$+18.2$	0.0036	$+16.1$	0.060
	PER	-46.8	$\mathbf{4.5 \times 10^{-8}}$	-49.4	$\mathbf{3.3 \times 10^{-5}}$
	n-step	-12.1	0.048	-14.7	0.11
	Avg.	$+22.6$	$\mathbf{0.0014}$	$+17.3$	0.15

Δ and p_t as in Table III; bold: $p_t \leq 0.05/16$. DDQN: Double DQN; Avg.: Averaged DQN. These endpoints are exploratory.

APPENDIX B
LEARNING-RATE REGRESSION

Table VI gives, for every arm and game, the ordinary least-squares regression of log AUC on the six sampled hyperparameters, each coded as a categorical factor with its first value as the reference (11 coefficients, 19 residual degrees of freedom). The learning-rate ratio is the back-transformed coefficient of 10^{-3} relative to 10^{-4}; the partial R^2 is the share of the variance left by the other five factors that the learning rate explains.

TABLE VI
ADJUSTED LEARNING-RATE EFFECT ON LOG AUC

Game	Arm	LR ratio (95% CI)	Partial R^2	R^2
Breakout	DQN	$0.56\ [0.49, 0.63]$	0.83	0.87
	DDQN	$0.53\ [0.48, 0.58]$	0.92	0.94
	PER	$0.60\ [0.48, 0.74]$	0.58	0.74
	n-step	$0.55\ [0.45, 0.67]$	0.68	0.84
	Avg.	$0.55\ [0.48, 0.63]$	0.83	0.88
Asterix	DQN	$0.21\ [0.11, 0.40]$	0.58	0.82
	DDQN	$0.11\ [0.05, 0.22]$	0.68	0.84
	PER	$0.38\ [0.18, 0.81]$	0.28	0.65
	n-step	$1.08\ [0.74, 1.56]$	0.06	0.37
	Avg.	$0.18\ [0.09, 0.37]$	0.57	0.84
Freeway	DQN	$0.57\ [0.45, 0.73]$	0.59	0.86
	DDQN	$0.51\ [0.42, 0.62]$	0.74	0.93
	PER	$1.10\ [0.61, 2.01]$	0.03	0.60
	n-step	$4.12\ [2.31, 7.34]$	0.59	0.80
	Avg.	$0.72\ [0.56, 0.92]$	0.30	0.72
Space Invaders	DQN	$0.30\ [0.26, 0.35]$	0.93	0.96
	DDQN	$0.32\ [0.26, 0.39]$	0.87	0.94
	PER	$0.33\ [0.20, 0.54]$	0.55	0.83
	n-step	$0.31\ [0.23, 0.42]$	0.78	0.89
	Avg.	$0.27\ [0.22, 0.32]$	0.93	0.96

LR ratio: multiplicative change in AUC from a learning rate of 10^{-4} to 10^{-3}, adjusted for the other five hyperparameters, with 95% t-interval. Partial R^2: of the learning rate given the other factors. R^2: of the full model.